

AKSHAY VYAS

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First author at **AAAI 2026** and **NeurIPS 2026 (Spotlight, top 1% of 30,709 submissions)**. PhD candidate (graduating May 2027) working on **how we measure and trust learned systems**: human-anchored evaluation of metrics, LLM/VLM judges, and reward models under optimization pressure, and valid-by-construction guarantees for inference. Nearly 3 years of building production software at Swiggy, Microsoft, and Goldman Sachs.

EDUCATION

The University of Texas at Dallas, Richardson, TX — Ph.D., Computer Science · GPA 3.95/4.0 Jan 2023 – May 2027 (expected)
The University of Texas at Dallas, Richardson, TX — M.S., Computer Science · GPA 4.0/4.0 Aug 2021 – Aug 2023
IIIT Hyderabad, India — B.Tech., Electronics & Communication Engineering · GPA 3.55/4.0 Aug 2015 – May 2019

RESEARCH EXPERIENCE

Graduate Research Assistant · UT Dallas · Advisor: Professor Nicholas Ruozzi · NSF Grant #2327245 Jan 2023 – Present

Invariants as diagnostic probes for evaluation metrics | *AAAI 2026 (17.6% acceptance)*

- Formalized approximate task invariants and used them to test metric reliability without human evaluators.
- Three tasks trained from scratch: image colorization, German-English machine translation, image captioning.
- Seven baselines retrained with and without invariants: ChromaGAN, ColorFormer, DDColor, BiBERT, Bi-SimCut, SmallCap, ClipCap.
- Auxiliary losses from frozen SAN, MiDaS, BERT-NER, DistilBERT, and scene-graph nets; no new labels.
- FDD improves 3-33%, FID 7-16%, FBD 13-15%, FBD-4 5-13%; PSNR, BLEU-4, METEOR, CIDEr move under 2%.
- Showed DINOv2 features are far more invariant-sensitive than Inception, arguing for richer encoders.

Human vs. automatic-judge sensitivity | *In preparation*

- Measures which semantic properties humans insist on preserving and which metrics, LLM judges and reward models miss.

Certified neural upper bounds for discrete inference | *NeurIPS 2026 Spotlight (top 1%); follow-up under review*

- First learned MAP upper-bound predictor whose validity is enforced by architecture, not post-hoc calibration.
- Constrains network outputs to a provably feasible region of the LP dual, by construction.
- Holds for trained, untrained, and adversarial outputs; smooth dual objective, no LP solver or optimal labels needed.
- Warm starts tighten the JGLP dual gap by up to six orders of magnitude within 100 iterations, across 15 benchmark models.
- Test-time adaptation lands within $1e-7$ of the exact optimum on 99% of constrained queries.
- Follow-up (under review): monotone, calibrated upper bounds inside AND/OR branch-and-bound, 5-10x tighter than mini-bucket.

Procedural error understanding and AR task guidance | *NeurIPS 2024 D&B; DMLR @ ICML 2023*

- Co-built CaptainCook4D, a large-scale egocentric dataset of procedural activities with annotated errors.
- Benchmarked vision and language models for procedural error detection; 50+ citations to date.

Multimodal perception for AR task guidance (DARPA PTG, demoed at MIT 2023) | *IEEE VR 2024 Workshop*

- Built a Detectron2 kitchen-object detection pipeline with training, inference, and Firebase retrieval.
- Open-set detection with Grounding DINO, plus CLIP and PROTO-CLIP for class refinement.
- Real-time segmentation with SAM and MobileSAM; 3D spatial mapping for object localization in AR.
- Co-localization fusing HoloLens 2 depth with SIFT matching in a real-time ROS pipeline.

PUBLICATIONS

† first author; longer author lists truncated at A. Vyas.

A Novel Approach to Evaluating Evaluation Metrics for Multi-Output Structured Prediction

AAAI 2026

† A. Vyas, A. Pimienta, N. Ruozzi · 17.6% acceptance · Code: github.com/akshay25vyas/invariants-metric-eval

Neural Dual Bounds: Valid-by-Construction JGLP Warm-Starts for MAP and Constrained MAP

NeurIPS 2026 Spotlight

† A. Vyas, S. Arya, B. Malhotra, T. Rahman, V. Gogate, N. Ruozzi

Learning to Bound: Monotonic, Calibrated Neural Upper Bounds for MPE Inference

Under review

† A. Vyas, H. Dong, S. Arya, B. Malhotra, T. Rahman, V. Gogate, N. Ruozzi

CaptainCook4D: A Dataset for Understanding Errors in Procedural Activities

NeurIPS 2024

R. Peddi, S. Arya, B. Challa, L. Pallapothula, A. Vyas, et al. · Datasets & Benchmarks track

Put on Your Detective Hat: What's Wrong in This Video?

DMLR @ ICML 2023

R. Peddi, S. Arya, B. Challa, L. Pallapothula, A. Vyas, et al.

Predictive Task Guidance with AI in Augmented Reality

IEEE VR 2024 Workshop

B. Rheault, S. Arya, A. Vyas, et al.

Camera Model Identification Based on Local Co-Occurrence Features

IEEE SP Cup 2018

P. Parkhiya, N. Reddy, A. Vyas, et al.

INDUSTRY EXPERIENCE

Software Development Engineer · Swiggy (Bundl Technologies) · Bengaluru, India Jun 2019 – Aug 2021

- Owned high-level and low-level architecture of ingress service relaying customer orders into cloud kitchens.
- The service carried the live order path at one of India's largest food delivery platforms.
- Ran service-level testing for the Spring Boot service by containerizing external services and databases.
- Shipped a fraud-detection REST API for delivery executives using OTP confirmation and geo-signals.
- Stack: Java (Spring Boot), Golang, Kafka, Docker, AWS (Lambda, Step Functions, DynamoDB), MySQL.

Software Development Engineer Intern · Goldman Sachs · Dallas, TX Jun 2022 – Aug 2022

- Built an OCR pipeline converting PDF financial tables into structured data for a banker-facing portal.
- Shipped the dashboard front end with AG Grid for interactive review (React, JavaScript, Python).

Software Development Engineer Intern · Microsoft (Bing Places) · Hyderabad, India May 2018 – Jul 2018

- Trained an automated spam-business classifier over local and global Bing data pipelines.
- Engineered and de-biased features; benchmarked classifier precision and recall. (C#, Model Tracker, MySQL)

TECHNICAL SKILLS

Machine learning: PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, NumPy/SciPy, pandas, Matplotlib, OpenCV, CVXPY

Evaluation & alignment: LLM and VLM judges (Qwen2.5-VL, Llama-3.2-Vision, Gemini); reward models (ArmoRM, Skywork-Reward); best-of-N and MBR selection; reward overoptimization analysis; forced-choice preference elicitation and inter-annotator agreement.

Metrics: COMET / xCOMET, BLEU, chrF++, METEOR, BERTScore, CIDEr, SPICE, CLIPScore/RefCLIPScore, FID, FDD (DINOv2), CMMD, LPIPS, PSNR, Fréchet BERT Distance.

Vision & multimodal: Detectron2, SAM / MobileSAM, Grounding DINO, CLIP, DINOv2, ROS, OpenCV, HoloLens 2, Firebase.

Methods: Multi-Output Structured Prediction, multi-task and auxiliary-loss training, probabilistic graphical models, LP and Lagrangian duality, branch-and-bound and AND/OR search, amortized and variational inference.

Languages: Python, C++, Java, Golang, SQL, JavaScript, Bash.

Systems: Docker, Kafka, Redis, Spring Boot, Flask, SQLite, React, AWS (Lambda, Step Functions, EC2, S3, DynamoDB, RDS), Git, Linux, Prometheus/Grafana.

RESEARCH ENGINEERING

Human studies and evaluation

- Web platform for human raters: Flask/SQLite, drag-and-drop ranking, autosave, built-in attention checks.
- Mixed-effects models, clustered bootstrap, and permutation tests over trial-level data.

Model training

- Implemented five invariant losses against frozen nets: Dice on masks, SILog on depth, KL on NER and sentiment logits.
- Swept only the invariant weight, holding every baseline hyperparameter fixed so metric movement was attributable.
- Full training matrix on single A100/V100 GPUs across seven datasets and three modalities.

Solver integration

- Integrated a neural bound predictor into a C++ branch-and-bound solver, queried at every search node.
- Optimized the Python-to-C++ inference path to millisecond latency across millions of nodes per search.
- End-to-end PyTorch training with CVXPY/Clarabel exact-LP references across UAI benchmark families.

AWARDS, SERVICE & TEACHING

- MVP award, Kitchen Intelligence & Optimization team, Swiggy (2021); under 3% of engineers per cycle.
- **Reviewer:** NeurIPS 2026.
- **Teaching Assistant, UT Dallas:** CS 6347 Statistical Methods in AI & ML; CS 4375 Introduction to Machine Learning, CS 6364 Artificial Intelligence, CS 4365 Intro to Artificial Intelligence
- **Teaching Assistant, IIIT Hyderabad:** Electronic Workshop II.
- Gold Medal, Men's Singles Table Tennis, National Inter-College Sports Fest (India); Sports Captain, IIIT Hyderabad.